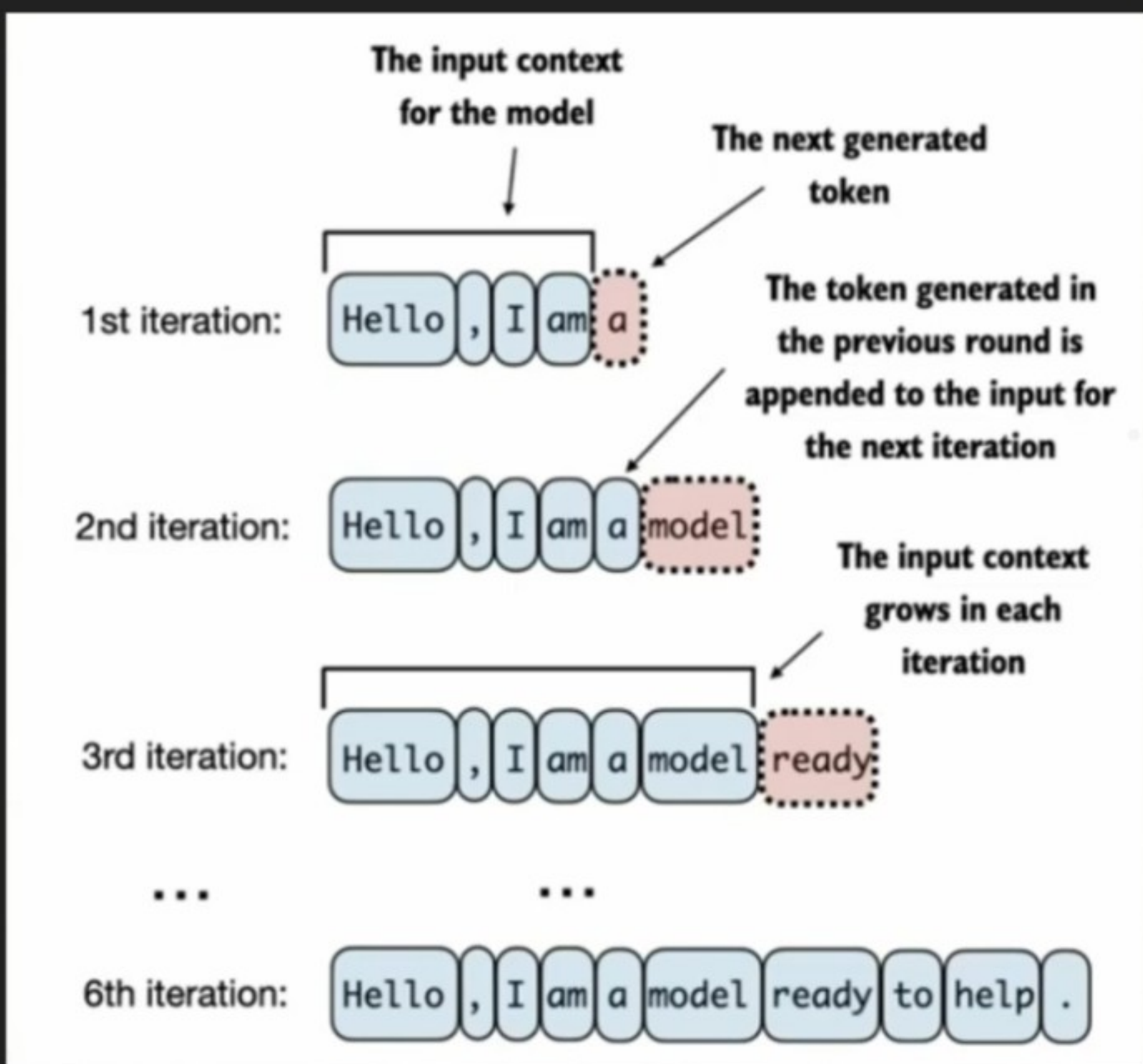


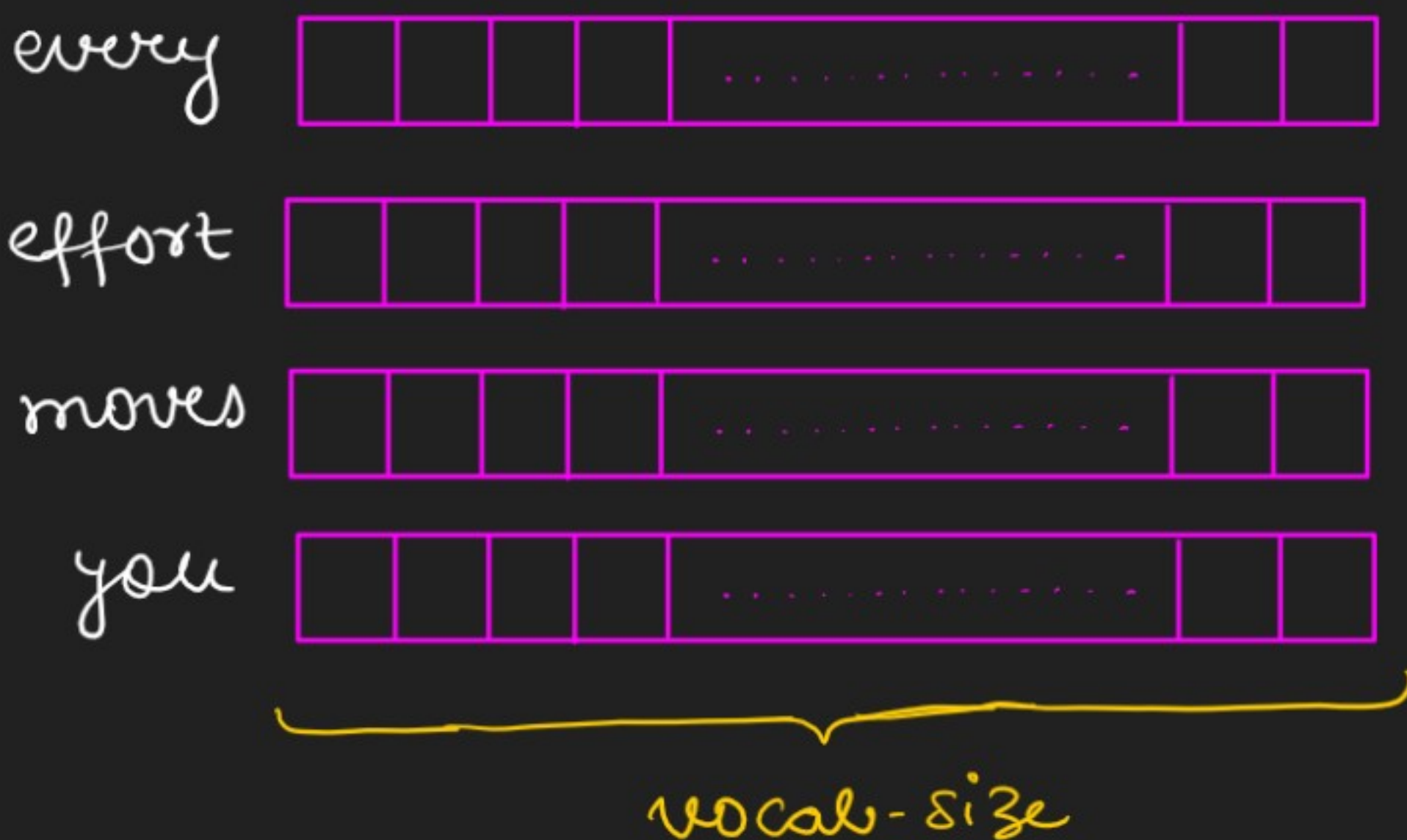
Lecture-25: Generating text

from output tensors

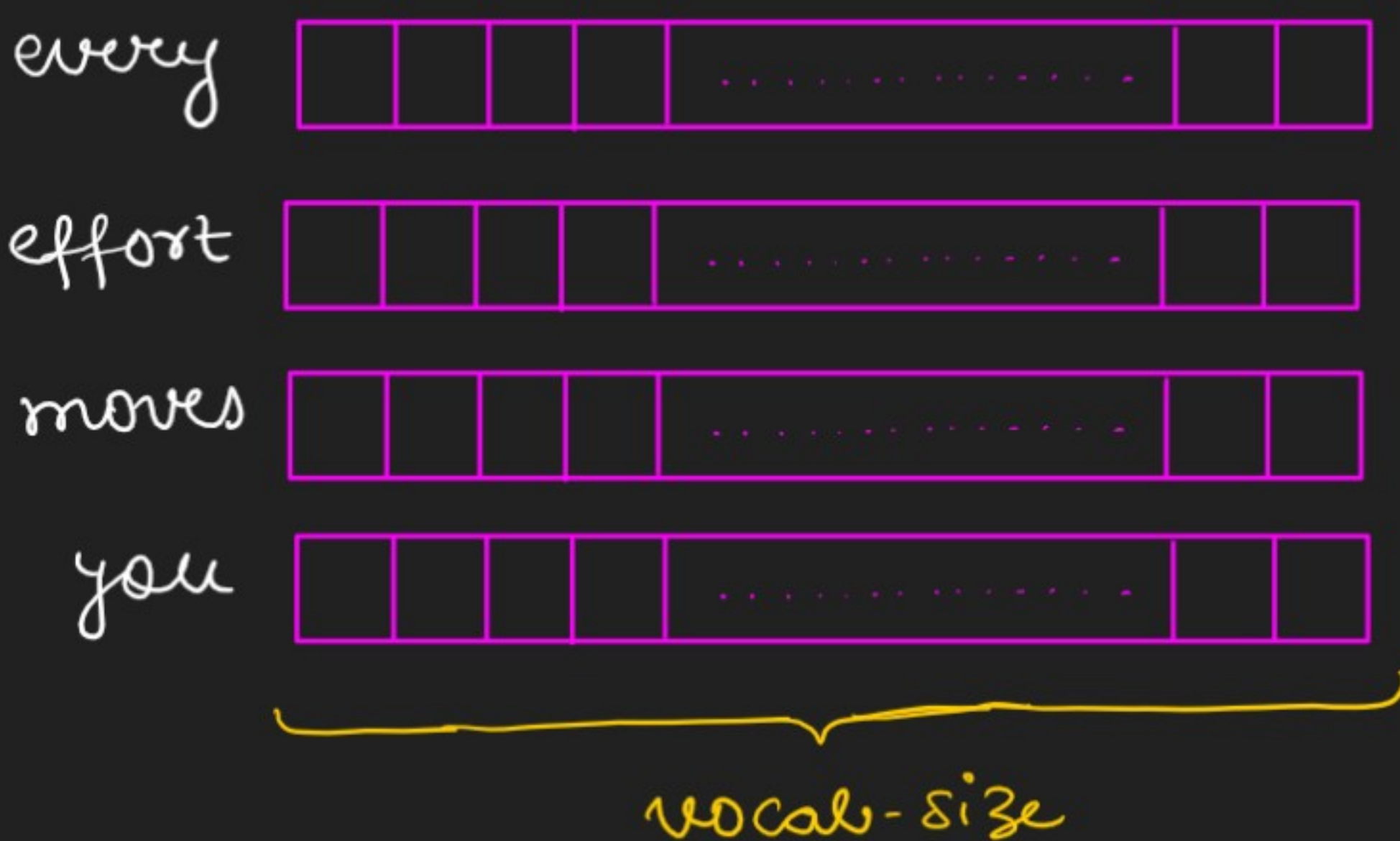
→ Review of how an LLM generate text



Step-1



Step-1



Step-2: Extract last vector



softmax

Step-3: convert logits into probability by applying softmax.

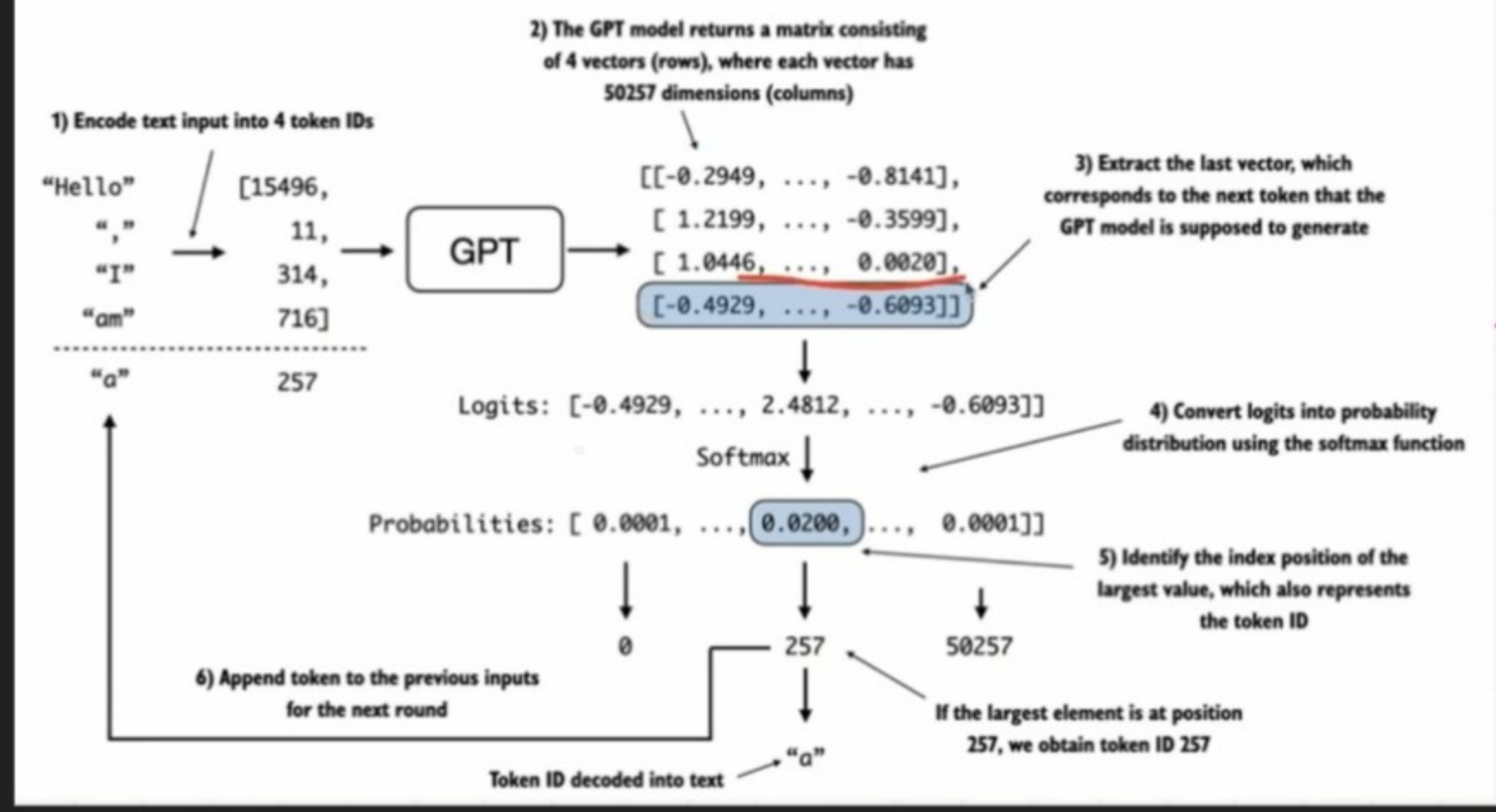


In the previous section:

↳ GPT model output tensor with shape.

[batch-size, num-token, vocab-size]

↳ How do we go from this to generated text.



* In each step, the model output a matrix with vectors representing the potential next token.

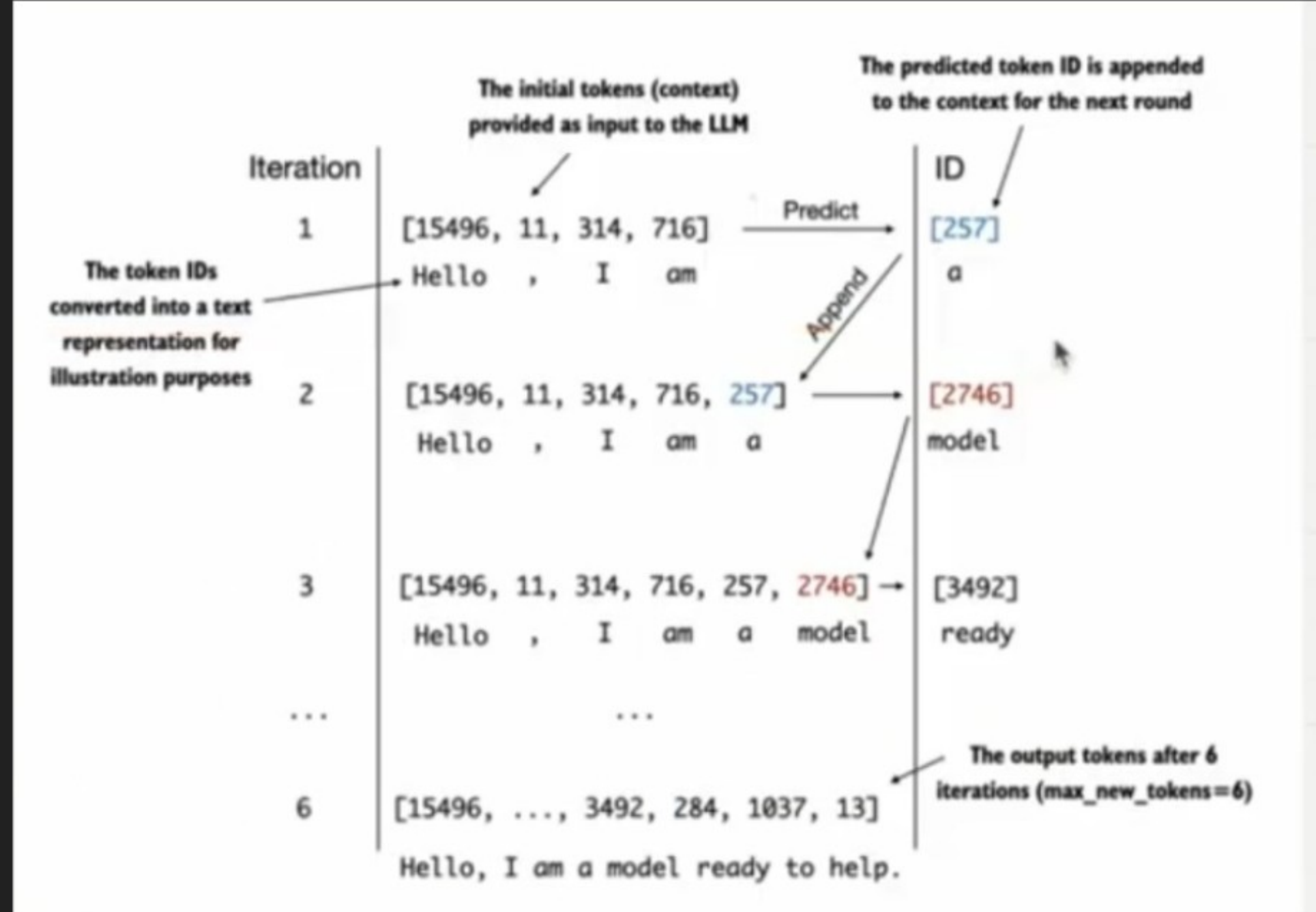
* The vector corresponding to the next token is extracted and converted into a probability distribution via the softmax function

index of highest value → Token ID
↓
Decode back to text
↓
Append to previous input ← produces next token input

This step-by-step process enables the model to generate text sequentially, building coherent phrases from the initial input context.

↳ In practice, we repeat this process over many iterations, until we reach a user specified no. of generated token.

The figure below illustrate the process of generating one token ID at a time.



In iteration-1, the model is provided with token corresponding to "Hello, I am"

predict next token with ID 257 ("a")

append to input

This process is repeated till the model produces the complete sentence

"Hello, I am a model

ready to help" after 6 iteration.